Reg. No									

St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore – 607001

PPH909 – QUANTUM MECHANICS - II

Time : 3 hrs **Max Marks :75**

SECTION – A (10X2=20)

Answer **ALL** Questions

1. Write down the equation for Heisenberg picture.
2. Give the applications of perturbation theory.
3. Define frame of reference.
4. What is the unit of cross section?
5. Define parity operator.
6. Differentiate bosons and fermions with example.
7. Define induced emission.
8. What are Einstein coefficients?
9. Define Hamiltonian.
10. Define quantization of field.

SECTION – B (5X5=25)

Answer **ALL** Questions

11. a) State and prove Fermi golden rule of transition to the space of continuum.

(or)

b) Explain Heisenberg picture.

12. a) Derive an expression for phase shift.

(or)

b) Discuss the validity conditions of Born approximation.

13. a) Derive an expression for distinguishability of identical particles.

(or)

b) Prove that parity operator commute with total Hamiltonian.

14. a) Give the semi classical theory of spontaneous emission.

(or)

b) Explain dipole approximation.

15. a) Derive the relativistic Hamiltonian equation.

(or)

b) Discuss the second quantization phenomenon.

SECTION –C (3X10=30)

Answer any **THREE** Questions

16. Explain (i) Schrodinger and (ii) Interaction pictures.

17. Using Born approximation derive the scattering cross section by a square well potential.

18. Describe the Hartee- Fock self consistent model.

19. Derive the Einstein's coefficients for induced and spontaneous emissions with the relation between them.

20. Explain the quantization of Electromagnetic field.

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